A APPENDIX A: TILT INVARIANCE AND NETWORK SPECIFICATIONS

1. Oblique catheter tilt invariance of the Stage 0 circular dark-core filter. Consider a cylindrical 6F plastic catheter of physical radius $r_c = 1.0$ mm inclined at an angle α relative to the axial slice normal $\hat{\mathbf{z}}$. Its intersection with an axial plane $z = z_0$ is an ellipse with semi-minor axis $a = r_c = 1.0$ mm perpendicular to the tilt projection and semi-major axis $b = r_c \sec\alpha$ along the tilt projection. For clinical perineal template and freehand insertions ($\alpha \leq 25^\circ$–30°), $\sec\alpha \leq 1.155$, so the in-plane dark core extends at most $b \leq 1.16$ mm from the centerline. Because Stage 0 evaluates six diameters across four physical radii $\mathcal{R} = \{1.35, 1.85, 2.50, 3.30\}$ mm and takes the outer radial maximum over the rank-2 and rank-3 directional order statistics,

$$S_{\text{ring}}(\mathbf{p}) = \max_{r\in\mathcal{R}} \left[0.45\, m_{(2),r} + 0.55\, m_{(3),r}\right],$$

any sampling radius $r \geq 1.85$ mm clears the obliquely elongated dark core along all six diameters for $\alpha \leq 30^\circ$, while tolerating one adjacent touching catheter ($m_{(1),r}$). When slice thickness is coarse ($\Delta z \in [4.0, 6.0]$ mm) and through-plane partial-volume averaging shifts the apparent dark core across $\Delta z \tan\alpha$ within the slice, initial cubic z-upsampling to 1.6 mm spacing restores axial continuity prior to Stage 0 evaluation.

2. `SynthBrachy3T` physical K-space and bundle simulation parameters. To pre-train the Stage 2B bundle graph transformer without clinical bundle annotations, `SynthBrachy3T` generates 300 realistic candidate bundles per fold. Each synthetic implant samples $N_{\text{true}} \sim \mathcal{U}\{12, 32\}$ physical catheters entering through a 60×50 mm^2 perineal template grid around a central vaginal obturator ($R_{\text{obt}} \sim \mathcal{U}[10, 15]$ mm), with correlated 3D quadratic deflections, 15%–25% touching twin pairs (≤ 3.0 mm clearance), 20% short obturator guide channels, random 1-to-3-slice mid-shaft signal dropouts, and $N_{\text{FP}} \sim \mathcal{U}\{6, 22\}$ vascular and fascial false-positive trajectories sampled at higher peripheral curvature and peripheral radial distances.

3. Summary of neural network hyperparameters across Stages 1 and 2.

- **Stage 1 (`NeedleUNet3D`):** 6 input channels, 4 resolution scales with `[24, 48, 96, 144]` channels, squeeze-and-excitation residual blocks, in-plane-only (1, 2, 2) pooling at the first two scales, 4 output heads ($P_{\text{mask}}, P_{\text{peak}}, V_x, V_y, P_{\text{obt}}$). Pre-trained for 45 epochs on 51 external pelvic MRI volumes (AdamW, learning rate 10^{-3}, weight decay 10^{-4}) and fine-tuned for 60 epochs per inner fold on dual-rater consensus traces.
- **Stage 2A (`NeedleSliceMPNN` + Collinear Stitcher):** 16-D node features, 10-D directed edge features across $\Delta k \in \{1, 2, 3, 4\}$ slices (≤ 24 mm), 4 triplet message-passing layers of hidden width 64 with cosine bending-angle attention; collinear stitcher bridges axial gaps ≤ 20 mm with midpoint error ≤ 2.2 mm and Δslope ≤ 0.14.
- **Stage 2B (`TipAwareBundleGraphTransformer`):** 26-D trajectory node tokens, 8-D pairwise geometric interaction edge features, 2 edge-biased local graph attention layers + 2 global bundle self-attention layers (width 96, 4 heads), temperature-calibrated on the 9 inner-validation patients per fold.

– **Stage 2C (`Dedicated1DTipLocalizer`):** 91 axial steps at 0.5 mm spacing over [−25.0, +20.0] mm along the cranial tangent, 22 channels (24 in 4d), 3 dilated 1D residual blocks ($d \in \{1, 2, 4\}$, kernel size 5, width 64) + 4-head 1D self-attention, trained with KL-divergence stopping loss and smooth-L_1 sub-slice offset regression, calibrated via split-conformal quantiles $\hat{q}_{1-\alpha}$ on the inner validation split.

B APPENDIX B: STAGE-BY-STAGE ABLATION AND ERROR AUDIT

Table 5 reports the complete 12-row stage-by-stage architectural ablation ladder across all 54 patients (1,063 catheters), on the 759 full-depth active-dwell shafts, and on the 23-patient severe catheter-adhesion subgroup, together with the ground-truth mask oracle upper bound for Xiao *et al.* (2026a).

TABLE 5 **Controlled architectural ablation and stage-by-stage progression on all 54 BWH patients (1 063 catheters, strictly nested 5 × 5 out-of-fold).** Traces the sequential addition of each mechanism from Stage 0 analytical dark-core symmetry through Stage 2C 1D tip profiling and literature fusion, alongside the oracle ground-truth mask upper bound.

	Detection, 1-to-1 (%)			E_{shaft} (mm)			E_{tip} (mm)		Time (s)
Method / Stage Configuration	Rec.	Prec.	F1	Mean	Median	P95	Mean	Median	CPU
Sequential stage-by-stage progression (whole cohort, 1 063 catheters)									
Stage 0: Dark-core prior (`k_lim_auto`)	82.8	69.1	75.3	0.60	0.47	1.41	9.91	3.08	**0.27**
Stage 1B: SSL + Multi-Rater U-Net + tubes	**94.2**	61.5	74.4	**0.57**	**0.46**	1.31	9.29	2.95	2.84
Stage 2A: Triplet MPNN only	93.9	73.5	82.5	0.58	0.47	1.31	8.95	2.83	4.09
+ Stage 2B: Bundle transformer + calibration	90.9	83.6	87.1	**0.57**	**0.46**	**1.30**	7.39	2.70	4.12
+ Stage 2B: Multi-rater consensus supervision	91.7	81.7	86.4	0.58	**0.46**	1.32	7.38	2.57	4.12
+ Stage 2A/B: Collinear split-fragment stitch	91.4	86.9	89.1	0.59	0.47	1.36	6.92	2.56	4.13
+ Stage 2C: 1D Tip Localizer (τ = 35 %)	93.6	83.4	88.2	0.59	0.47	1.37	6.71	2.32	4.15
+ Stage 2C: 1D Tip Localizer (τ = 50 %, 4b)	91.4	86.9	89.1	0.59	0.47	1.37	**6.23**	2.27	4.15
+ Stage 2C: 1D Tip Localizer (τ = 55 %)	91.3	**87.6**	**89.4**	0.59	0.47	1.37	6.33	2.27	4.15
+ Lit. Fusion (Dai TV + Kostoulas + Kim, 4d)[a]	91.4	87.4	**89.4**	0.58	0.47	1.36	**6.23**	**2.26**	4.18
Xiao 2026 on GT masks (oracle upper bound)[b]	97.7[r]	83.5[r]	90.0[r]	0.20[r]	0.11[r]	0.83[r]	8.94[r]	1.68[r]	0.12[r]

[a] Combines Kim *et al.* (2026) inverse-variance sub-voxel Poly3 fitting, Dai *et al.* (2020) axial TV continuity + Aleong *et al.* (2024) 5-model epistemic uncertainty features, and Kostoulas *et al.* (2025) iterative multi-segment gap-bridging.
[b] Oracle upper-bound check following Xiao *et al.* (2026) Table 2 Part B (tracker run directly on ground-truth binary masks). Source: `2026-10-23-unified-clean-nested-benchmark.json` and `2026-10-24-peer-reviewed-baselines-and-fusion.json`.

1. Itemized stage-by-stage progression analysis:

– **Stage 0 alone (physical dark-core filter):** Using zero learned parameters, Stage 0 with automatic template cropping already finds 82.8% of all catheters (95.3% of long shafts) with 69.1% precision (**75.3% F1**, 0.47 mm median shaft error, 3.08 mm median / 9.91 mm mean tip error). However, it cannot distinguish round blood vessels from catheters.

- **Adding Stage 1B (self-supervised, multi-rater 3D U-Net):** Regressing sub-pixel centerline peaks and 3D directions boosts recall to **94.2%** (97.0% on long shafts) and cuts mean tip error to 9.29 mm (2.95 mm median), but because every candidate seed is still tracked greedily, each patient carries 11.6 false vascular tracks on average (61.5% precision, 74.4% F1).
- **Adding Stage 2A (triplet message-passing graph network):** Penalizing kinks across consecutive slice links eliminates 43% of false tracks while keeping **93.9% recall**, lifting precision by +12.0 percentage points to 73.5% (**82.5% F1**, 8.95 mm mean / 2.83 mm median tip error).
- **Adding Stage 2B (bundle graph transformer, multi-rater supervision, and collinear fragment stitching):** Evaluating the candidate bundle as a whole (87.1% F1 with temperature calibration; 86.4% F1 with multi-rater Stage 2 supervision) and stitching collinear fragments separated by 1-to-3-slice dropouts raises precision to **86.9%** and F1 to **89.1%** (91.4% recall) while reducing mean tip error to 6.92 mm (2.56 mm median).
- **Adding Stage 2C (dedicated 1D axial tip localizer, 4b):** Inspecting the 0.5 mm sub-slice axial profile at the cranial end of each trajectory cuts mean 3D tip error from 6.92 mm down to 6.23 **mm** and median tip error from 2.56 mm down to 2.27 **mm** (95th percentile reduced to 27.27 mm).
- **Adding Literature-Inspired Fusion (4d: sub-voxel inverse-variance cubic fitting + epistemic uncertainty & axial TV continuity + iterative multi-segment gap-bridging):** Inverse-variance-weighted sub-voxel cubic fitting (Aleong *et al.*, 2024; Kim *et al.*, 2026) reduces mean shaft error from 0.589 mm to 0.583 **mm** ($p = 1.84 \times 10^{-8}$; long-shaft median 0.45 → 0.44 mm); epistemic uncertainty & axial TV continuity features (Dai *et al.*, 2020; Aleong *et al.*, 2024) reduce median 3D tip error to 2.26 **mm** (axial tip median 1.99 → 1.94 mm, tips ≤ 3 mm 58.7% → 59.0%); and iterative multi-segment gap-bridging & polynomial split suppression (Kostoulas *et al.*, 2025) eliminate 6 residual split-fragment false alarms, raising precision to **87.4%** and F1 to **89.4%** ($p = 0.028$).

2. Complete catheter-level root-cause audit of residual errors across all 54 patients:

- **What causes the 146 unmatched predictions (2.7 per patient)?** Nearly a third (**31.5%**, 46/146) are residual split or near-duplicate segments of an already-detected catheter, **20.5% (**30/146**)** correspond to real catheters supported by Expert Team 2 within 5.0 mm (and, among the 100 unmatched predictions on the 38 CT-paired patients, 52.0% correspond to a catheter on CT), and 2.7% (4/146) lie within 3.5–6.0 mm of an unmatched reference track. Only **45.2% (**66/146**, or 1.22 per patient)** are genuine background false alarms (> 6.0 mm from any reference catheter), occurring along dark pelvic vessels or fascial intersections.
- **What causes the 91 missed catheters (1.7 per patient)?** Before Stage 2B thresholding, our 1,960-track candidate pool contains **97.4% (**1,035/1,063**)** of all reference catheters within 3.5 mm (and **98.8%** [750/759] of the long active-dwell shafts). Fewer than a third of missed catheters (**30.8%,** 28/91) are lost upstream due to severe multi-slice signal dropouts. The remaining **69.2%**

(63/91) are traced by Stages 1 and 2A but filtered out in Stage 2B—most often because their bundle confidence falls just below the $\tau = 50\%$ threshold (46 catheters) or because they lose one-to-one matching inside a tight cluster of touching needles (15 catheters). Lowering τ to 35% recovers more than a quarter of these misses, raising whole-cohort recall to **93.6%** and long-shaft recall to **94.7%** (precision 83.4%, F1 88.2%).

– **Why do some tips have errors > 10 mm?** Decomposing every 3D tip error vector into its longitudinal component ($\Delta s_{\parallel}$) and lateral component ($\Delta r_{\perp}$) shows that **98.8% of total squared 3D tip error across all 972 matched catheters (and 99.3% in the > 10 mm tail) is purely longitudinal along the catheter track**, whereas lateral tip error is 0.72 mm median. When a tip error exceeds 10 mm (162 catheters, 16.7%), it is driven by mid-shaft signal dropout (41.4%), dark bowel/air pockets sitting directly atop the catheter tip (25.3%), intrinsic human inter-rater tip ambiguity > 6 mm (16.7%), crowded tip arrays (12.3%), or top-of-scan truncation (4.3%).

C APPENDIX C: SPECIFICITY AND MULTI-MODALITY LITERATURE

Table 6 details the 5-row zero-needle false-positive specificity study across all 48 public needle-free 3T T2W pelvic MRI volumes (`MSD Task05_Prostate` (Antonelli *et al.*, 2022)), and Table 7 compares NeedleFinder 2026 against published catheter digitization methods across CT, ultrasound, electromagnetic tracking, and MRI.

TABLE 6 **Zero-needle out-of-distribution specificity on 48 public clinical 3T pelvic T2W MRI volumes (MSD `Task05_Prostate`).** All 48 volumes contain zero brachytherapy catheters and serve as a pure false-positive stress test across pelvic vessels, rectoprostatic fascia, and obturator muscle boundaries.

	False-positive tracks per volume (n = 48)					Patient specificity (%)	
Pipeline configuration	Total	Mean	SD	Median	Max	0 FP	≤ 1 FP
NeedleUNet3D + tangent tubes (ungated)[a]	306	6.38	4.77	6.0	19	14.6	18.8
Unpruned Stage 0 + Stage 1 pool[b]	294	6.12	4.70	5.0	23	4.2	10.4
Ungated Unified **NeedleFinder 2026** (τ = 50 %)[c]	197	4.10	4.45	3.0	23	16.7	27.1
Ungated Unified **NeedleFinder 2026** (τ = 60 %)[c]	171	3.56	3.48	3.0	17	16.7	29.2
Stage 1 U-Net + Stage 0 radial gate[d]	18	0.38	0.53	**0.0**	2	64.6	97.9
Frozen Unified **NeedleFinder 2026** + Dark-Core Gate (τ = 50 %)[e]	16	0.33	0.75	**0.0**	3	79.2	91.7
Clean SSL + Multi-Rater Stage 1 + Stage 0 gate[f]	**6**	**0.125**	**0.33**	**0.0**	**1**	**87.5**	**100.0**

[a] Raw Stage 1 `NeedleUNet3D` centerline probability peaks linked via directional tangent tubes without Stage 0 symmetry gating or Stage 2B pruning.

[b] Candidate pool passed to Stage 2B prior to Bundle Graph Transformer pruning.

[c] Unified **NeedleFinder 2026** pipeline evaluated without the Stage 0 dark-core gate at τ = 50 % (4.10 FP/scan) and τ = 60 % (3.56 FP/scan).

[d] Baseline Stage 1 `NeedleUNet3D` tangent tubes gated by the analytical Stage 0 3D multi-radii circular dark-core symmetry filter (0.38 FP/scan).

[e] Full Unified **NeedleFinder 2026** pipeline with Stage 0 dark-core symmetry gating at τ = 50 % (16 total FPs = 0.333/scan; 38/48 [79.2 %] 0-FP scans; 44/48 [91.7 %] ≤ 1-FP scans).

[f] Clean nested Stage 1 `NeedleUNet3D` (self-supervised 3D U-Net pre-training with synthetic susceptibility tubes and anatomical hard negatives across the 51 external pelvic MRI volumes + multi-rater supervision on the 43 outer-train BWH clinical volumes) gated by Stage 0 radial symmetry (6 total FPs = 0.125/scan; 42/48 [87.5 %] 0-FP scans; 48/48 [100.0 %] ≤ 1-FP scans). Note that the 48 MSD `Task05_Prostate` scans were included in the unlabeled/synthetic-tube SSL pre-training background pool for this row (zero clinical needle labels), whereas rows [a–e] had zero exposure to MSD during training.

TABLE 7 **Published automatic catheter and needle reconstruction studies in brachytherapy (2013–2026), grouped by imaging modality.** Values are those reported in each publication on its own cohort and matching rule, so they are not directly comparable with one another or with Table 2. Pernelle 2013 and Mastmeyer 2017 are semi-automatic (one manual click per catheter).

Study	Modality	Site	Pts	Cath.	Rec./Prec. (%)	E_{shaft}	E_{tip}	Time Stat.
CT								
Weishaupt *et al.* 2022	CT	Prostate	57[l]	353[a]	–	0.13	–	~60 s mean
Xie *et al.* 2023	CT	GYN post-op	70	210	–[b]	–	–	– Dice / HD95
Wang *et al.* 2024	CT	GYN IC/IS	17	56[o]	–	1.8	1.1	– mean
Quetin *et al.* 2025	CT	Breast	117	408[a]	–	0.70	1.37	118 s mean
Xiao *et al.* 2026[m]	CT	GYN recurrent	38	203[o]	91.1 / 90.7	0.31	0.88	11.6 s mean
Xiao *et al.* 2026 (JBHI)	CT	Pelvic seeds	100		– F1 90.3	–	1.05	9.4 s mean tip
Ultrasound								
Zhang *et al.* 2020	TRUS	Prostate	23	339	98 / –	0.09	0.33	<1 s mean
Andersén *et al.* 2020	TRUS	Prostate	1102[l]	7207[l]	100 / –	0.55[c]	–	– median RMS
Liu *et al.* 2022	TRUS	Prostate	242	3500	80[d] / –	–	–	– frac. ≤ 2 mm
Goulet *et al.* 2025	3D US	Prostate	102[o]		– 93.4 / –	0.44	–	– mean
MRI								
Pernelle *et al.* 2013	3T T2W	GYN IC/IS	10	101	92.1 / –	–[e]	–	0.3 s/cath. frac. < 2 mm
Mastmeyer *et al.* 2017	3T T2W	GYN IC/IS	54	760	93[f] / –	1.49[g]	–	3 s/cath. mean HD
Zaffino *et al.* 2019	3T T2W	GYN IC/IS	50	826	98.5 / 93.3[h]	2.0	–	2–9 min mean dist.
Dai *et al.* 2020	MRI	Prostate	20	299	–	0.93	0.37[n]	– mean / signed
Shaaer *et al.* 2022	1.5T + markers	GYN IS	20	360	100 / –	0.97	–	11 min[i] mean shift
Aleong *et al.* 2024	MRI	Prostate	35	214[o]	95 / –	–	–	0.9 s –
Kostoulas *et al.* 2025	MRI	Prostate	34[o]	261	100 / 100	0.75[j]	1.07[j]	– median
Kim *et al.* 2026	T2W	GYN hybrid	30	35[o]	98.5 / –	0.74	2.52	37 s mean
NeedleFinder 2026 (all 1 063)	3T T2W	GYN IC/IS	54	1063	91.4 / 86.9	0.47[j]	2.27[j]	**4.15 s** median
NeedleFinder 2026 (759 long)	3T T2W	GYN IC/IS	54	759	91.3 / 88.7[k]	0.45[j]	2.41[j]	**4.15 s** median

[a] Test-set catheters.
[b] Reported as Dice 0.88–0.90 and HD_{95} 0.71–0.77 mm.
[c] Root-mean-square distance to the clinical reconstruction.
[d] Fraction of catheters within 2 mm along 90 % of their length.
[e] Maximum Hausdorff distance below 2 mm for 92 % of catheters.
[f] Fraction within 3.4 mm.
[g] Mean Hausdorff distance.
[h] Cluster-level: 1.5 % false negatives and 6.7 % false positives among 826 catheters.
[i] Assisted reconstruction time, versus 46 min manual.
[j] Median (means: 0.59 ± 0.40 mm shaft and 6.23 ± 10.49 mm tip on all 1 063 catheters; 0.57 ± 0.39 mm shaft and 7.41 ± 12.23 mm tip on the 759 long shafts).
[k] Strict one-to-one recall at $\tau = 50$ % (92.5 % with inner-OOF cross-fit; 94.7 % at $\tau = 35$ %); precision counts predictions matched to short obturator tracks as correct (90.0 % $F1_{adj}$, 90.5 % with inner-OOF cross-fit). Times for **NeedleFinder 2026** are full-volume uncached per-patient averages across all 54 scans on CPU (4.15 s overall: 1.26 s on the 20 standard 320 × 256 scans, 5.85 s on the 34 high-resolution 640 × 416 / 768 × 768 scans) and one RTX 4090 GPU (2.67 s overall: 0.67 s standard, 3.85 s high-resolution).
[l] Treatments, not patients; 7207 of 24422 needles formed the evaluation set.
[m] Fully supervised model; their 10-shot synthetic-to-real variant reports 89.4 / 89.0, 0.33 mm and 0.90 mm.
[n] Signed along-axis tip displacement (mean ± SD 0.37 ± 1.68 mm), not an absolute 3D distance.
[o] Reported in plans (Wang 2024: 56 plans), fractions (Xiao 2026: 203 fx), scans (Goulet 2025: 102 scans; Kostoulas 2025: 34 scans), or volumes (Aleong 2024: 214 vol.; Kim 2026: 35 vol.) rather than individual catheters/patients.

D APPENDIX D: EXTENDED DOSIMETRY AND GENERALIZATION TABLES

Tables 8 and 9 provide all 11 cross-organ, multi-sequence, 4D intra-procedural insertion, and paired Pelvic CT generalization configurations, together with the complete protocol-by-protocol AAPM TG-43U1 ^{192}Ir dosimetric distributions (including bladder, rectum, and sigmoid proxy D_{2cc} differences), simulated clinician tip-review curves ($k \in \{0, 1, 2, 3, 5, 8, 10\}$), and split-conformal coverage at 80%–95% nominal levels.

TABLE 8 **Zero-shot transfer across organs, secondary MRI pulse sequences (including T1 VIBE failure boundary), 4D serial insertion, and paired post-implant Pelvic CT (Part C supporting Table 4).**

Part C: Cross-organ, seq., 4D & CT[a]	n_{vol}	n_{cath}	Rec. %	Prec. %	F1 %	Sh. mm	Tip mm	Key finding
Prostate 3T: Stage 0 (`k_lim_auto=40`)	1	14	**100.0**	87.5	93.3	0.34	2.09	14/14, 2 FPs
Prostate 3T: Zero-Exp. No-SSL (`auto=40`, τ=50)	1	14	92.9	**100.0**	96.3	0.33	**0.99**	13/14, 0 FPs
Prostate 3T: Zero-Exp. No-SSL (`ACSV=47`, τ=50)	1	14	**100.0**	**100.0**	**100.0**	0.34	1.57	14/14, 0 FPs
Prostate 3T: Stage 1 Full Nested (`auto=40`)	1	14	**100.0**	**100.0**	**100.0**	0.30	1.08	14/14, 0 FPs
Prostate 3T: Stage 0 zeroed ($X_{1:3}$ = 0, `auto=40`)	1	14	0.0	–	0.0	–	–	985× drop
Prostate 3T: 1-ch ResUNet + clDice (`auto=40`)	1	14	**100.0**	56.0	71.8	**0.24**	2.48	11 vascular FPs
3D wbSSFP / True-FISP ($\Delta z \leq 2.5$ mm)	2	47	**97.9**	88.5	**92.9**	0.72	3.40	vs. 10.62 PMB19
Thick-slice 2D AX T2 ($\Delta z \leq 6.0$ mm)	49	882	88.4	67.9	76.9	1.07	4.89	vs. 57.6 % 1-ch
Post-contrast AX T1 VIBE ($\Delta z \leq 4.0$ mm)	12	217	58.5	53.8	56.1	1.87	8.08	vs. 61.5 % Stage 0
4D Serial MRI: Indep. 3D (Cases 68–71)	23	108	88.0	76.0	81.6	0.45	2.71	4-pt pooled
4D Serial MRI: Causal Kalman (Cases 68–71)	23	108	77.8	**88.4**	82.8	0.75	3.57	+12.4 % prec.
Paired Pelvic CT (38 paired / 40 vols, ≤ 5 mm)	38	801	79.2	–	–	2.01	2.78	52.0 % extra conf.

[a] 'Sh. mm' and 'Tip mm' report median in-plane shaft and 3D tip errors (mm). 'Zero-Exp. No-SSL' denotes the 5-fold GYN ensemble trained with zero prostate exposure (excluding Case 100 from SSL pre-training; `auto=40` uses automatic `k_lim_auto` slice 40, whereas `ACSV=47` uses the clinical template crop slice 47). On 'Thick-slice 2D AX T2', 882 is the number of reference catheters exhibiting a physical dark-core signal in the thick-slice slab (out of 1 019 geometrically intersecting). In 4D monitoring (4 patients, Cases 68–71, 23 serial scans, 1.31 s mean monitoring runtime, each evaluated strictly with its held-out outer-fold model), per-case final-scan F1s (Independent 3D → Causal 4D Kalman) are Case 68: 77.2 % → 83.0 %, Case 69: 84.7 % → 82.7 % (precision 83.7 % → 93.9 %), Case 70: 85.7 % → 93.3 %, and Case 71: 76.2 % → 66.7 % (precision 72.7 % → 100.0 %); mean tip error improves from 11.23 → 9.75 mm. Persisted catheters within ≤ 10 mm of newly inserted needles exhibit 1.44 mm median (1.61 mm mean) lateral shift vs. 1.10 mm at 10–20 mm and 1.00 mm (1.19 mm mean) at > 20 mm (Mann–Whitney $p = 9.97 \times 10^{-4}$ for ≤ 12 mm vs. > 12 mm); depth deflection reaches 3.29 mm median (4.56 mm mean, 10.49 mm P90) at > 90 mm depth ($\rho = 0.555$, $p = 5.55 \times 10^{-9}$). In Pelvic CT, 38 of 40 scanned CT volumes pair with the 54-case MRI benchmark (801 MRI-paired reference catheters evaluated for cross-modality correspondence across non-rigid inter-scan deformation: 68.9 % matched at ≤ 3.5 mm, 79.2 % at ≤ 5.0 mm; 52/100 [52.0 %] extra MRI detections vs. `CY` confirmed on CT).

TABLE 9 **Dosimetric translation and simulated clinician tip triage (Parts A & B; complete breakdown supporting Table 4).** Part A evaluates AAPM TG-43U1 HDR ^{192}Ir dosimetric sensitivity (7.0 Gy prescription per fraction, 5.0 mm dwell step spacing) under Protocol A (reference dwell times transferred to predicted catheters) and Protocol B (dwell times re-optimized on predicted catheters, dose evaluated on reference geometry). Part B evaluates simulated uncertainty-guided 1-click tip relocation ($\hat{\sigma}_{tip}$) and unreviewed ($k = 0$) nested 5-fold split-conformal 3D tip prediction intervals across the $n = 972$ matched out-of-fold catheters.

	Dwell-wt. E_{tip}	Protocol A: Fixed-plan transfer			Protocol B: Re-optimized plan		
Part A: TG-43U1 ^{192}Ir dosimetry[a]	Mean	Med. $\lvert\Delta D_{90}\rvert$ (%)	$\gamma_{2/2}$ (%)	$\gamma_{3/3}$ (%)	$\lvert\Delta D_{90}\rvert$ (%)	Cran. D_{2cc}	$\gamma_{2/2}$ (%)
PMB 2019 3D U-Net + DBSCAN (n=54)	10.51	8.36 27.6 / 27.2	64.0	73.3	37.1 / 10.2	95.3 %	43.8
Stage 0: 3D radial symmetry (n=54)	9.04	7.51 23.0 / 21.3	68.2	76.5	17.8 / 8.9	54.0 %	48.6
NeedleFinder 2026 w/o Synth (4a, n=54)	6.65	5.49 18.5 / 13.2	73.4	81.5	12.8 / 5.8	35.2 %	59.5
NeedleFinder 2026 proposed (4b, n=54)[b]	5.72	4.66 15.2 / 11.5	76.6	84.3	8.2 / 4.2	29.5 %	66.0
NeedleFinder 2026 + Top-2 Tip Review (n=54)	**4.56**	**3.93 11.7 / 10.2**	**80.2**	**87.3**	**5.9 / 2.8**	**22.5 %**	**74.3**
Human Rater 2 (`LB/AMe`, n=40)	6.27	5.54 21.6 / 16.3	68.1	77.4	20.9 / 4.0	57.7 %	59.3
NeedleFinder 2026 proposed (4b, n=40)[b]	5.18	3.93 13.5 / 11.1	79.0	86.4	7.0 / 3.2	21.1 %	71.6
NeedleFinder 2026 + Top-2 Tip Review (n=40)	**4.12**	**3.33 10.5 / 10.1**	**81.9**	**88.7**	**5.1 / 2.6**	**17.3 %**	**78.2**

Part B1: Simulated $\hat{\sigma}_{tip}$ tip triage (n = 972)[c]	Reloc.	Mean	Med.	P90	P95	≤ 2 mm	≤ 3 mm	> 10 mm
Unreviewed **NeedleFinder 2026** (k=0 tips/pt)	0	6.23	2.27	16.20	27.27	45.6 %	58.7 %	16.7 %
Simulated tip review (k=1/pt, ~8 s)	35	5.54	2.06	14.60	23.54	48.6 %	61.9 %	14.4 %
Simulated tip review (k=2/pt, ~16 s)	69	5.07	1.95	12.58	21.48	51.3 %	64.8 %	12.8 %
Simulated tip review (k=3/pt, ~24 s)	105	4.57	1.81	10.91	19.16	54.0 %	67.2 %	11.1 %
Simulated tip review (k=5/pt, ~40 s)	169	3.72	1.67	7.79	14.26	59.5 %	72.6 %	8.0 %
Simulated tip review (k=8/pt, ~64 s)	251	2.77	1.51	5.98	9.90	65.5 %	78.5 %	5.0 %
Simulated tip review (k=10/pt, ~80 s)	301	**2.52**	**1.44**	**4.97**	**8.30**	**69.2 %**	**81.8 %**	**4.3 %**

Part B2: Split-conformal (k=0, n=972)[c]	Nom.	Emp.	Fixed	Mean r	Med. r	Fixed r	Low-$\hat{\sigma}$ r	Width red.
Conformal 80 % 3D tip interval (k=0)	80.0 %	79.3 %	80.2 %	7.31	6.46	7.73	4.46	−16.4 %
Conformal 85 % 3D tip interval (k=0)	85.0 %	84.4 %	85.5 %	9.80	8.71	11.43	5.96	−23.8 %
Conformal 90 % 3D tip interval (k=0)	90.0 %	**90.2 %**	89.9 %	13.71	12.11	17.68	8.41	−31.5 %
Conformal 95 % 3D tip interval (k=0)	95.0 %	**95.4 %**	95.2 %	21.05	18.42	28.39	12.89	−35.1 %

[a] TG-43U1 evaluation uses 5.0 mm dwell step spacing (3.5 mm distal tip offset, ≤ 45 mm active length) on a 1.5 mm isotropic 3D dose grid with implant-conformal HR-CTV$_{implant}$ (8.0 mm radial envelope, D_{90} = 7.0 Gy) and 2 cc OAR proxy shells. $\lvert\Delta D_{90}\rvert$ cells report mean / median percentage target dose shift; 'Cran. D_{2cc}' reports mean relative error $\lvert\Delta D_{2cc}\rvert$ (%) in the cranial (sigmoid/fundal) OAR proxy shell; $\gamma_{2/2}$ and $\gamma_{3/3}$ are global 3D gamma pass rates (2 %/2 mm and 3 %/3 mm). Dwell-weighted tip error $E_{tip}^{dwell\text{-}w}$ correlates with $\lvert\Delta D_{90}\rvert$ at Spearman $\rho = 0.8629$ ($p = 4.9 \times 10^{-17}$) and with $\gamma_{2\%/2mm}$ at $\rho = -0.8474$ ($p = 6.5 \times 10^{-16}$).

[b] Paired Wilcoxon vs. PMB 2019 ($p = 4.9 \times 10^{-6}$ on $\lvert\Delta D_{90}\rvert$, $p = 8.2 \times 10^{-7}$ on $\gamma_{2\%/2mm}$), vs. Stage 0 ($p = 7.9 \times 10^{-8}$ on $\lvert\Delta D_{90}\rvert$, $p = 1.4 \times 10^{-7}$ on $\gamma_{2\%/2mm}$), and vs. Human Rater 2 on $n = 40$ ($p = 0.0278$ on $\lvert\Delta D_{90}\rvert$, $p = 0.0109$ on $\gamma_{2\%/2mm}$).

[c] Part B1 simulates a retrospective 1-click distal tip relocation oracle ($E_{tip} \rightarrow \sqrt{\Delta r_{\perp}^2 + \min(\lvert\Delta s_{\parallel}\rvert, 0.85\,\text{mm})^2}$ when $E_{tip} > 2.0$ mm, E_{shaft} unchanged; 'Reloc.' = total tips relocated across 54 patients) ranked by out-of-fold $\hat{\sigma}_{tip}$. Part B2 reports unreviewed ($k = 0$) 5-fold split-conformal 3D tip prediction interval half-widths (radii r in mm) across the $n = 972$ matched OOF catheters ('Fixed': unscaled residual quantile; 'Low-$\hat{\sigma}$ r': median half-width on the lower-uncertainty half of catheters; 'Width red.': adaptive vs. fixed median half-width reduction). Simulated oracle count-guided recovery (0.65 clicks/patient) recovers +17 missed catheters (972 → 989/1 063, 93.0 % recall).